Peihan Liu

New York • +1 7405017352 • peihanliu@cs.columbia.edu

EDUCATION

Columbia University New York, NY

- Ph.D. student in Computer Science 2029 (expected)
- Advisors: Rachel Cummings, Roxana Geambasu

Harvard University Cambridge, MA

- M.Eng. in Computational Science and Engineering 2024
- Thesis: Multicalibration and Loss Minimization; Advisor: Cynthia Dwork

University of Michigan Ann Arbor, MI

- B.S. in Mathematics and Statistics with high distinction and high honors 2022
- Math REU at UM (2021 summer); Advisor: Martin Strauss

RESEARCH EXPERIENCE

Student Researcher, Google Research 05/2025-01/2026

Supervisor: Alex Bie, Lillian Tsai, Natalia Ponomareva (Google Research, MAD)

- Developed *ContinuousBench*, a continuously regenerated benchmark for evaluating capability gains from DP synthetic text, including corpus/QA curation and continued-pretraining receipt on Gemma 3 models.
- Developed *PATH*, a DP tabular data synthesis framework leveraging autoregressive LLMs to preserve temporal dependencies. Evaluated Gemma 3 models, achieving over 60% reduction in trajectory distributional distance compared to leading marginal-based methods.

Research Intern, OpenDP 05/2023-08/2024

Supervisor: Salil Vadhan, Wanrong Zhang (OpenDP/Harvard)

- Developed optimal numerical compositors for sequential DP composition; identified privacy-accounting failures caused by inaccurate privacy curve approximations and introduced realizable curves with conservative preprocessing guarantees.
- Developed *DP-CLIP* for private vision-language representation learning; fine-tuned CLIP/BLIP on vision-language tasks and established convergence and privacy–utility trade-offs.

TEACHING EXPERIENCE

- Teaching Assistant, COMS6998: Private Systems, Columbia (Roxana Geambasu) 2025
- Teaching Fellow, CS226R: Algorithmic Fairness, Harvard (Cynthia Dwork) 2024

PUBLICATIONS

General ML:

1. M. Afzali, H. Ashtiani, A. Bie, G. Kamath, **P. Liu**, A. Mouzakis, V. Singha. *Actually, All Learnable Distribution Classes Are Privately Learnable.* Submitted, 2026.
2. V. Sriraman, **P. Liu**, D. J. Hsu, A. Block. *Behavior Cloning is Not All You Need: The Optimality of On-Policy Distillation for Noisy Expert Feedback.* NeurIPS, 2026 (Spotlight).

   Workshops: DEMO @ ICML 2026, FoPt @ COLT 2026
3. S. Yang, **P. Liu**, C. Pehlevan. *Convex Relaxation for Solving Large-Margin Classifiers in Hyperbolic Space.* TMLR, 2024.

Privacy & Fairness:

4. **P. Liu**, L. Rosenblatt, W. Kong, N. Ponomareva, G. Kamath, R. Cummings, R. Geambasu, Y. Gan, L. Tsai, A. Bie. *ContinuousBench: Can differentially private synthetic data transfer knowledge?* Submitted, 2026

Workshops: TPDP 2026 (Oral), Re-Data @ COLM 2026 (Oral), MemFM @ ICML 2026

5. L. Rosenblatt, **P. Liu**. *A Natively Differentially Private Foundation Model from Noisy Chebyshev Moments*. Submitted, 2026
6. **P. Liu**, A. Caulfield, M. Chen, R. Cummings, R. Geambasu, M. Lécuyer, P. Tholoniat. *Adaptive Target-Charging with Privacy Filters and Individual Accounting.* NeurIPS, 2026
7. L. Rosenblatt, **P. Liu**, R. McKenna, N. Ponomareva. *Privately Fine-Tuned LLMs Preserve Temporal Dynamics in Tabular Data.* ICML, 2026

Workshops: TPDP 2026 (Oral)

8. A. Huang, **P. Liu**, R. Nakada, L. Zhang, W. Zhang. *Safeguarding Data in Multimodal AI: A Differentially Private Approach to CLIP Training.* Preprint, 2024.
9. Z. Zhang, **P. Liu**, X. Wang, C. Xu. *Improving Adversarial Transferability with Scheduled Step Size and Dual Example.* Preprint, 2023.

Earlier Works on Algebraic Coding Theory:

10. H. Ou-azzou, M. Najmeddine, N. Aydin, **P. Liu**, B. Ialou, E.M. Mouloua. *Linear codes invariant under a linear endomorphism.* Advances in Mathematics of Communications, 2023.
11. F. Caliskan, R. Aksoy, N. Aydin, **P. Liu**. *Entanglement-Assisted Binary Quantum Codes from Skew Cyclic Codes over F2 × (F2 + vF2).* Quantum Information Processing, 2023.
12. A. S. Karbaski, T. Abualrub, N, Aydin, **P. Liu**. *Additive polycyclic codes over F4 induced by nonbinary polynomials.* Journal of Applied Mathematics and Computing, 2023
13. B. Caliskan, N. Aydin, **P. Liu**. *Skew cyclic codes over Z4 + u Z4 + vZ4.* Cryptography and Communications, 2023.
14. A. Bellil, K. Guenda, N. Aydin, **P. Liu**, T. A. Gulliver. *Constacyclic and quasi-twisted codes over Zq[u]/⟨u2−1⟩ and new Z4-linear codes.* Advances in Mathematics of Communications, 2023.
15. A. S. Karbaski, T. Abualrub, N, Aydin, P. Liu. *Additive Polycyclic Codes over F4 Induced by Binary Vectors and Some Optimal Codes.* Advances in Mathematics of Communications, 2021.
16. N. Aydin, **P. Liu**, B. Yoshino. *Polycyclic Codes Associated with Trinomials: Good Codes and Open Questions.* Design, Codes, and Cryptography, 2021.

FELLOWSHIPS & AWARDS

OpenMined Fellowship	2026-2027
Outstanding Achievement in Mathematics Award	2022
M. S. Keeler Department of Mathematics Merit Scholarship	2021
University Honor & James B. Angell Scholar	2020-2022

SERVICE

Reviewer for: NeurIPS, ICML, ICLR, AISTATS, COLT, RANDOM, SATML, AAAI, Quantum Inf. Process., IEEE Trans. Inf. Theory., JACODES, DMAA